Malawi, step 2 (2026-11-02)

chain	seed	iterations.saved	acceptance.rate	final.log.likelihood	best.log.likelihood	errored.proposals
1	20263016	5000	0.279	−36889.7	−36873.4	0
2	20263017	5000	0.279	−36819.5	−36807.9	4
3	20263018	5000	0.277	−36858.9	−36832.2	0
4	20263019	5000	0.279	−36871.0	−36853.0	0

file
mcmc_malawi_step2_chain1_2026-10-31.Rdata
mcmc_malawi_step2_chain2_2026-10-31.Rdata
mcmc_malawi_step2_chain3_2026-11-01.Rdata
mcmc_malawi_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):

trate.3	age.50.and.over.transmission.multiplier.3	trate.4
3.000	2.627	2.572
age.20.to.29.transmission.multiplier.0	age.20.to.29.transmission.multiplier.3	age.20.to.29.transmission.multiplier.2
2.179	2.084	2.029
unsuppressed.disengagement.rates	trate.0	suppressed.disengagement.rates
1.902	1.883	1.864
male.engagement.multiplier	trate.1	age.15.to.19.transmission.multiplier.3
1.821	1.811	1.799
trate.2	log.OR.engagement.slope	male.suppression.multiplier
1.759	1.742	1.700
female.to.male.multiplier	log.OR.testing.slope	age.20.to.29.transmission.multiplier.1
1.691	1.671	1.598
unsuppression.rates	age.30.to.39.transmission.multiplier.0	age.30.to.39.transmission.multiplier.2
1.586	1.571	1.548
age.15.to.19.transmission.multiplier.0	age.0.to.4.hiv.mortality.multiplier.0	age.30.to.39.transmission.multiplier.3
1.544	1.537	1.494
age.20.to.24.base.aging.rate		
1.448		

Merged simset:

location	simulations	total.weight	file
Malawi	1000	1000	simset_malawi_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000–2025): 54 of 63 targets flagged (worst 30 shown; all in <location>_fit_check.csv)

data.type	age	sex	years.with.data	pct.years.in.sim.95	median.sim.vs.data	last.year	last.sim.vs.data
incidence	10–19	female	26	8	x0.51	2025	x0.43
hiv.mortality	25–49	male	26	0	x0.53	2025	x0.04
hiv.mortality	50 and over	female	26	0	x1.83	2025	x1.45
hiv.mortality	50 and over	both	26	4	x1.83	2025	x1.38
hiv.mortality	50 and over	male	26	12	x1.79	2025	x1.30
incidence	10–19	both	26	15	x0.57	2025	x0.50
hiv.mortality	25–49	both	26	0	x0.61	2025	x0.03
incidence	10–19	male	26	15	x1.63	2025	x1.01
hiv.mortality	25–49	female	26	8	x0.66	2025	x0.03
awareness	0–14	both	11	0	+15.9 pts	2025	+9.0 pts
hiv.mortality	10–19	female	26	15	x1.36	2025	x0.88
hiv.mortality	10–19	both	26	15	x1.27	2025	x0.57
suppression	0–14	both	11	9	+9.9 pts	2025	+19.1 pts
hiv.mortality	All ages	male	26	8	x0.81	2025	x0.54
hiv.mortality	All ages	both	26	4	x0.84	2025	x0.54
awareness	15+	male	11	0	+6.5 pts	2025	+5.1 pts
hiv.mortality	15–24	female	26	46	x1.25	2025	x0.66

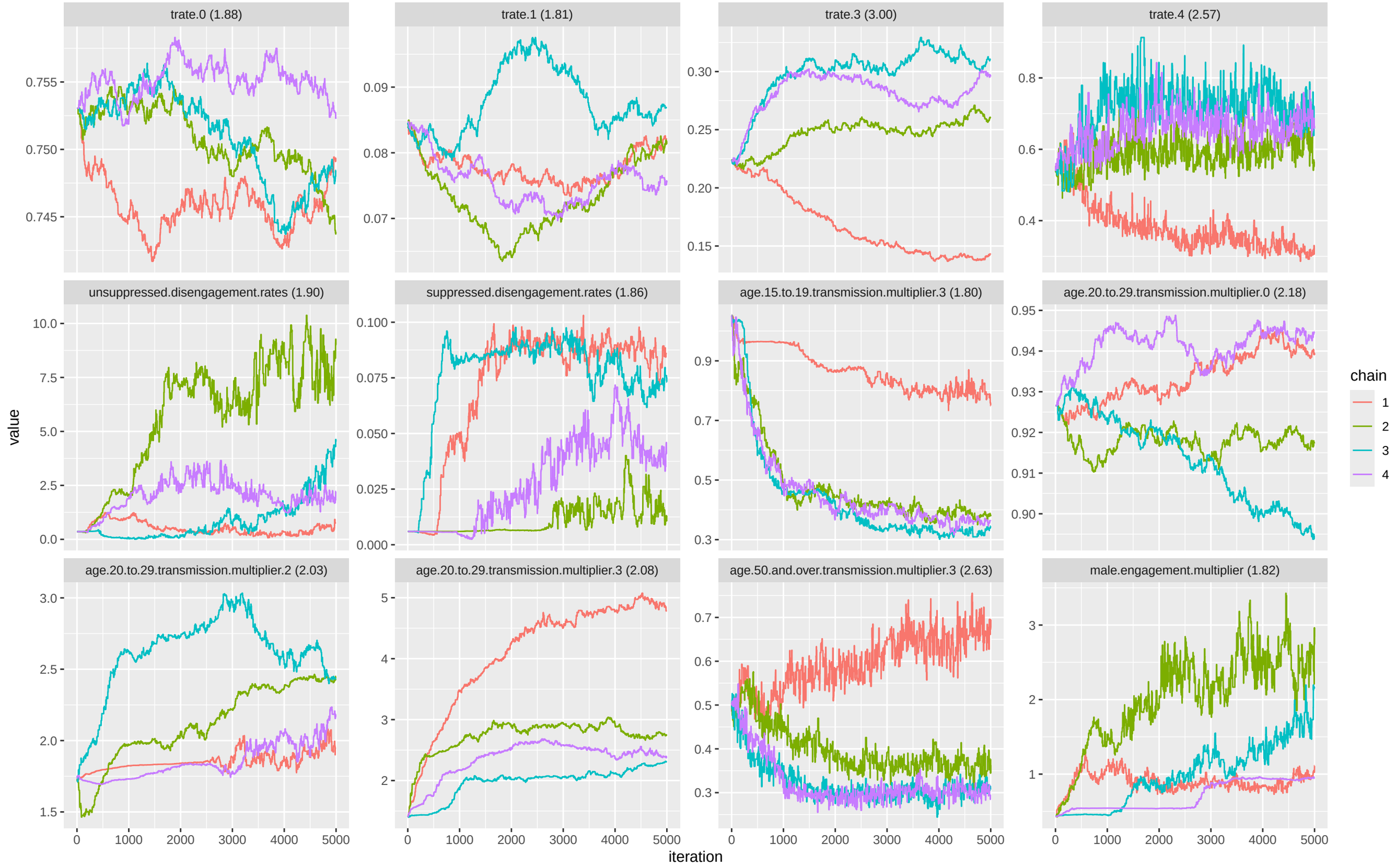
Malawi step 2: likelihood

0
−10000
−20000
−30000
−40000
value
0
25000
50000
75000
100000
iteration

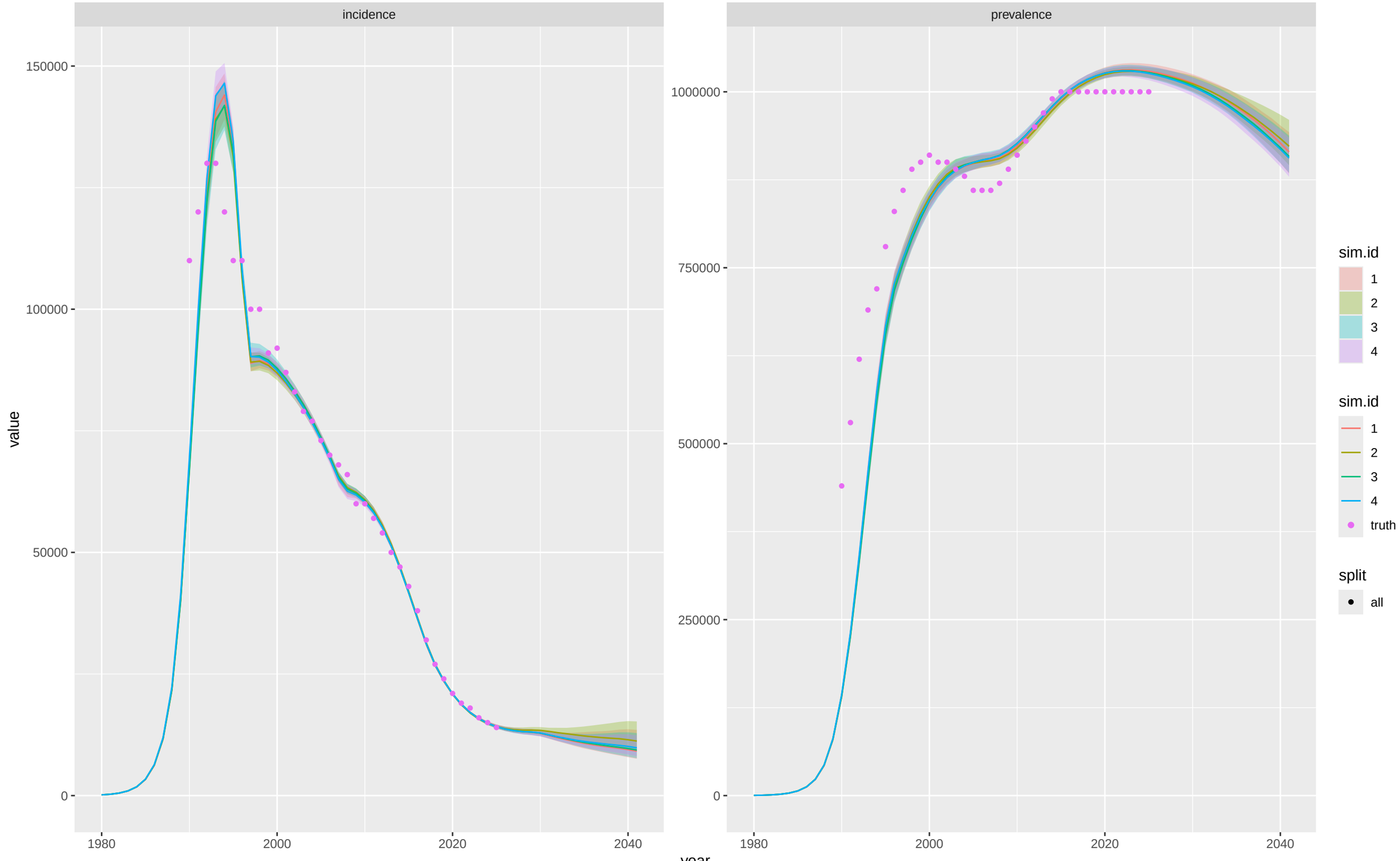
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

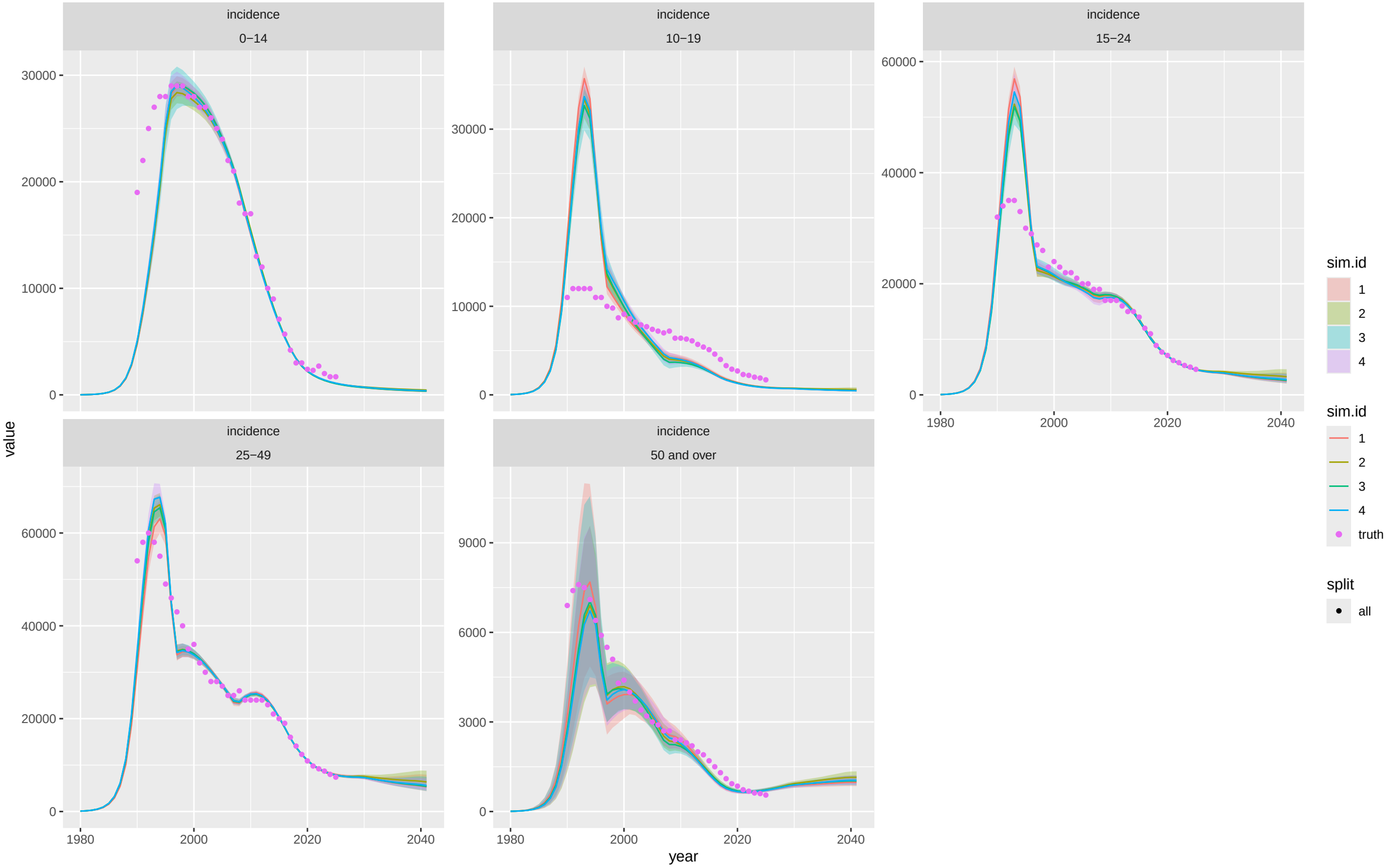
Malawi step 2: trace



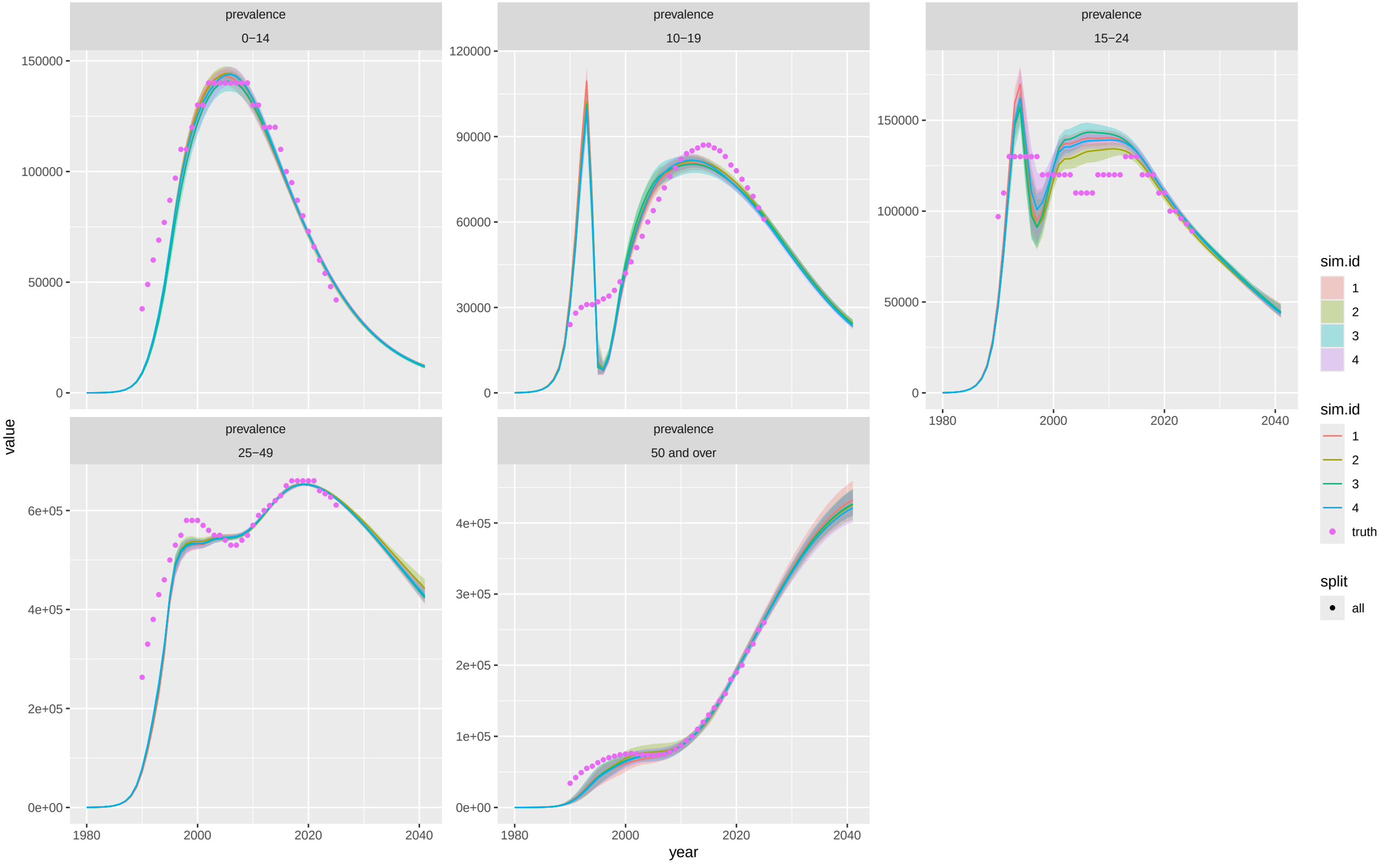
Malawi step 2: fit (one color per chain)



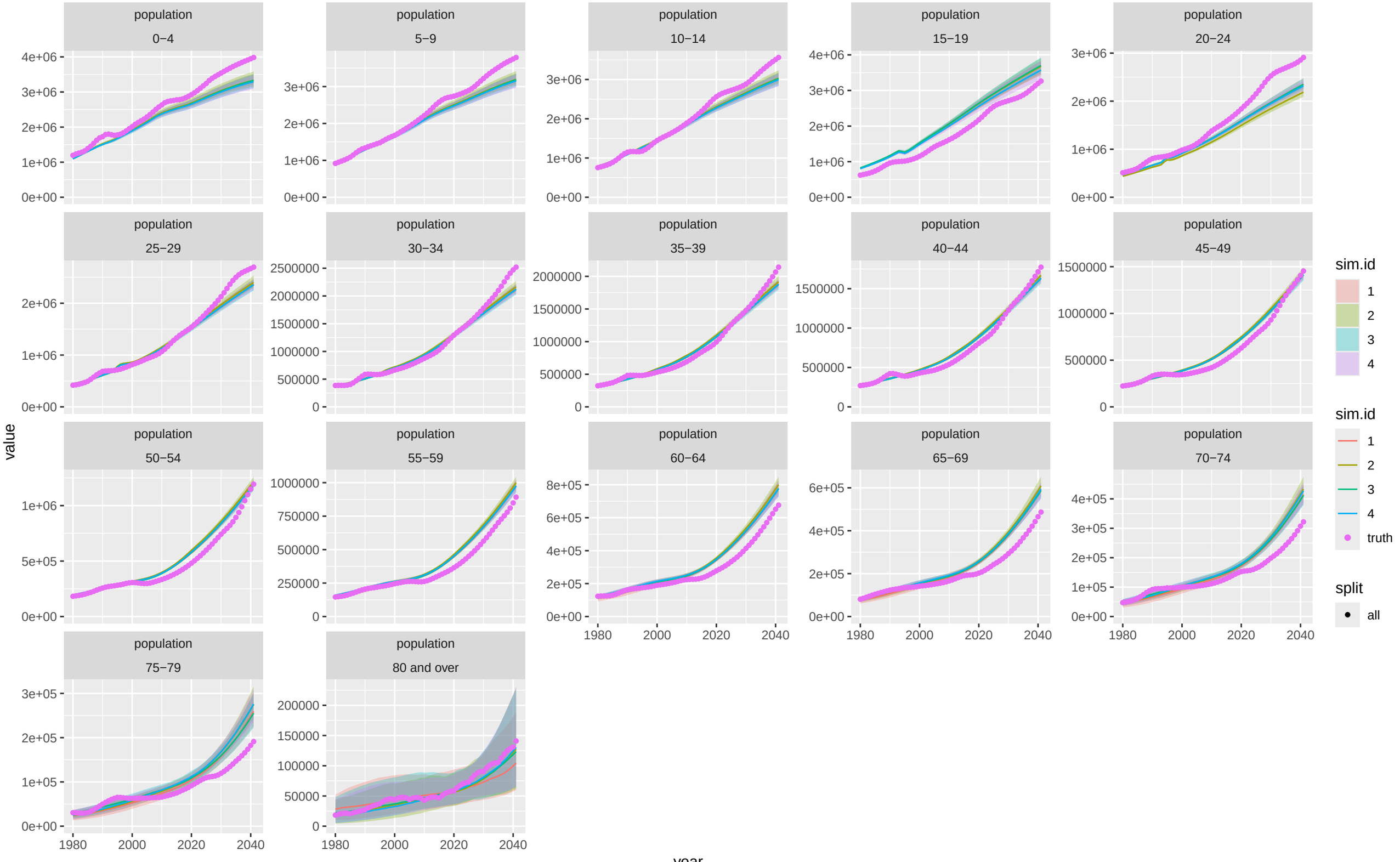
Malawi step 2: incidence.by.age (one color per chain)



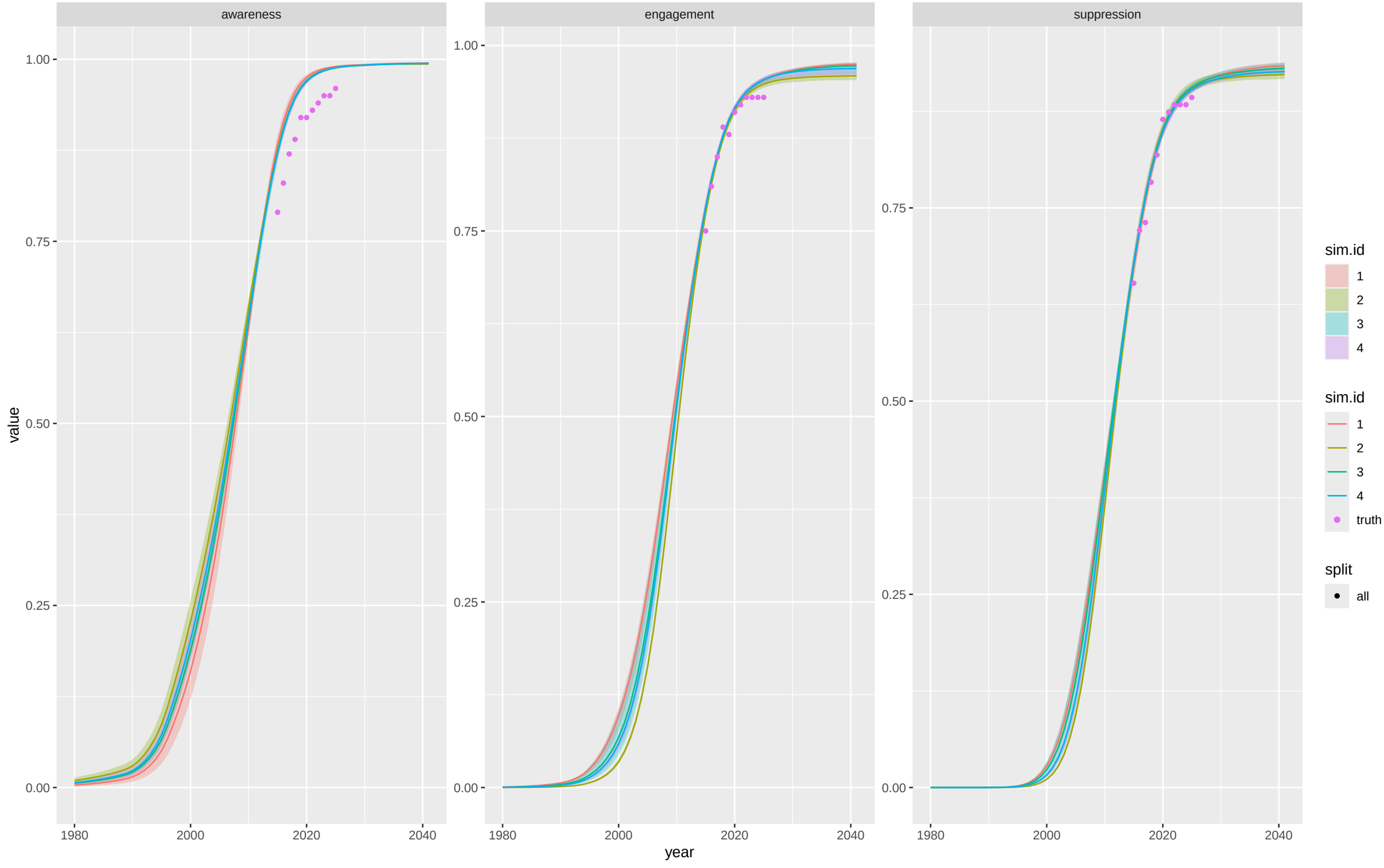
Malawi step 2: prevalence.by.age (one color per chain)



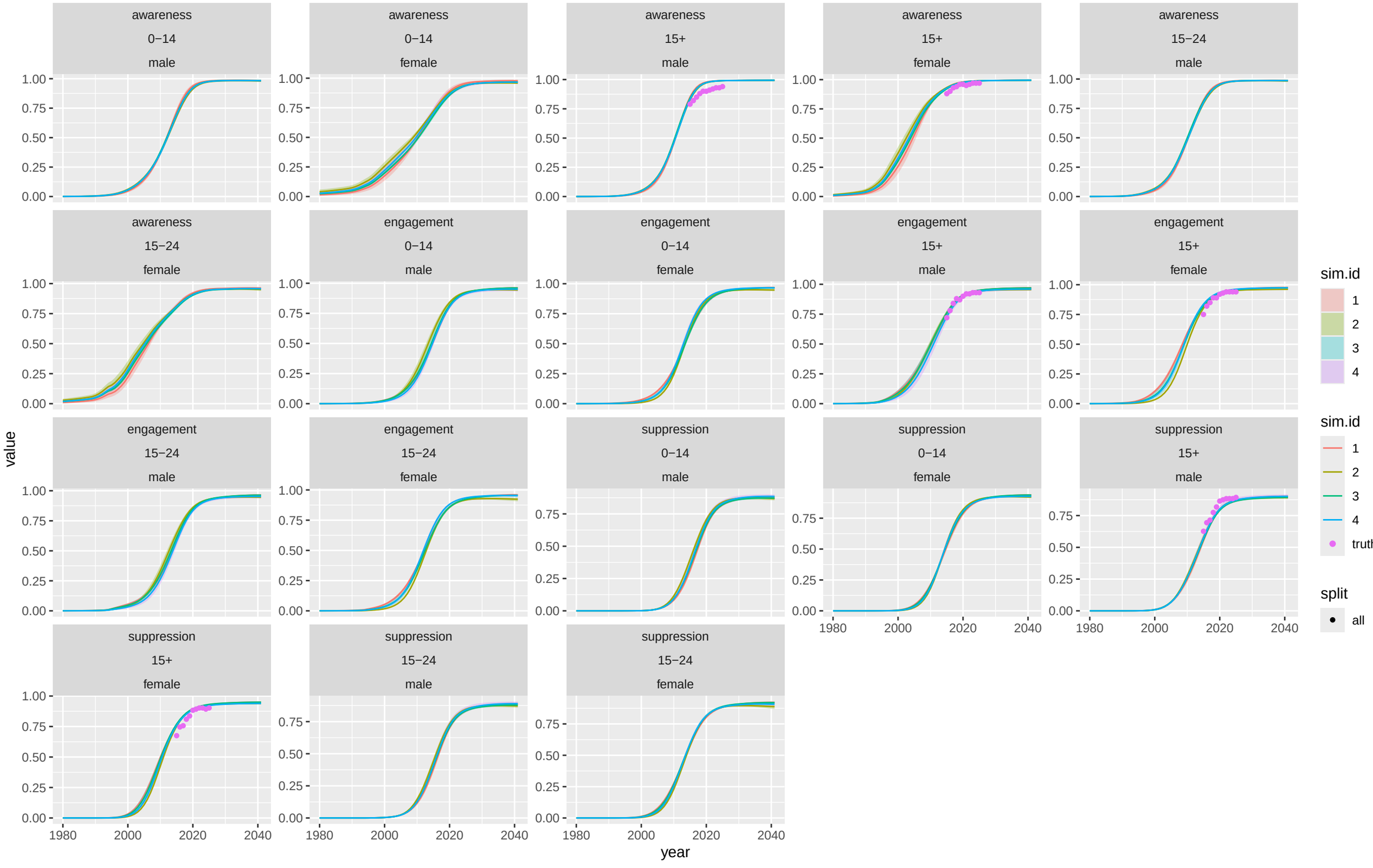
Malawi step 2: population.by.age (one color per chain)



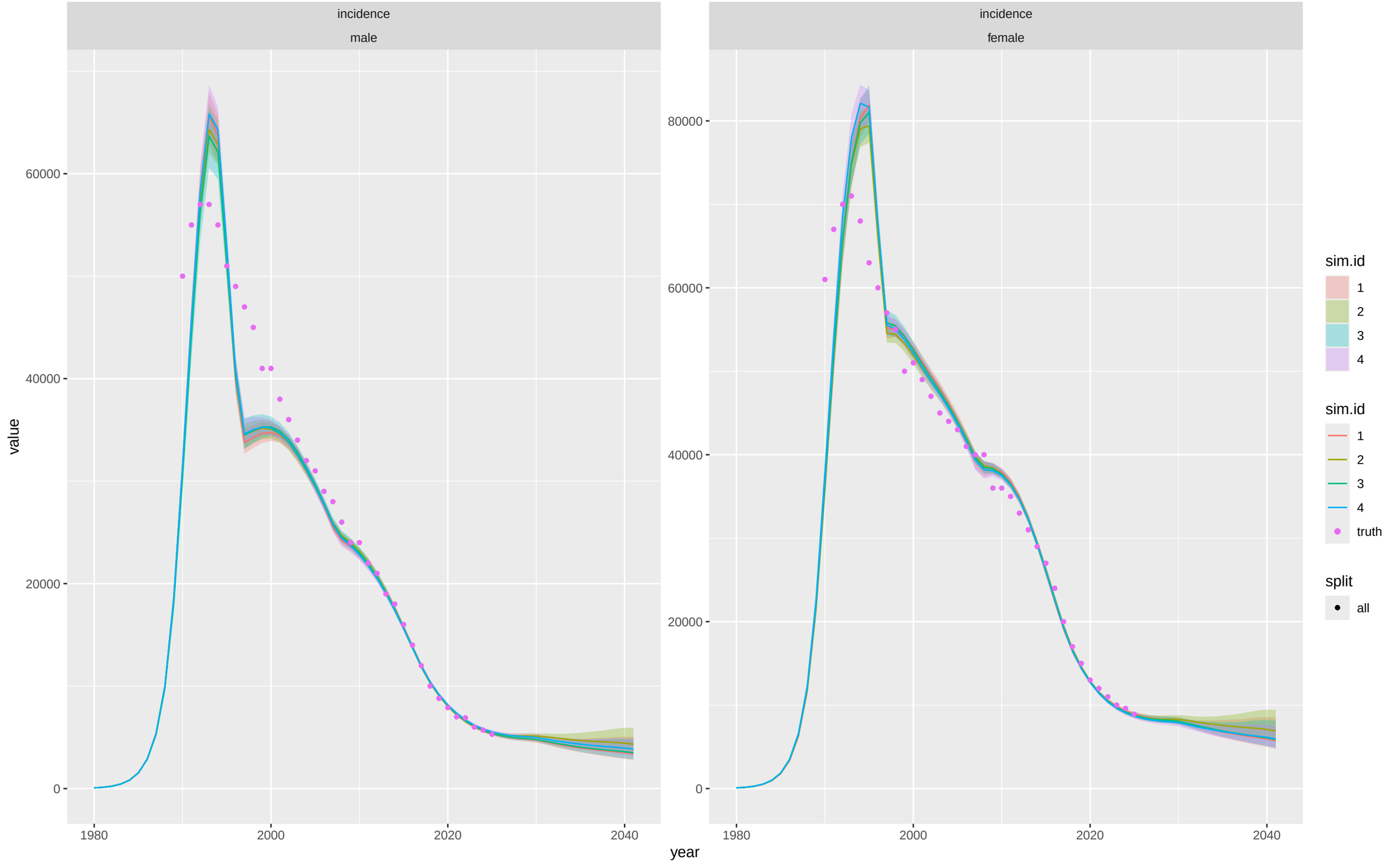
Malawi step 2: cascade (one color per chain)



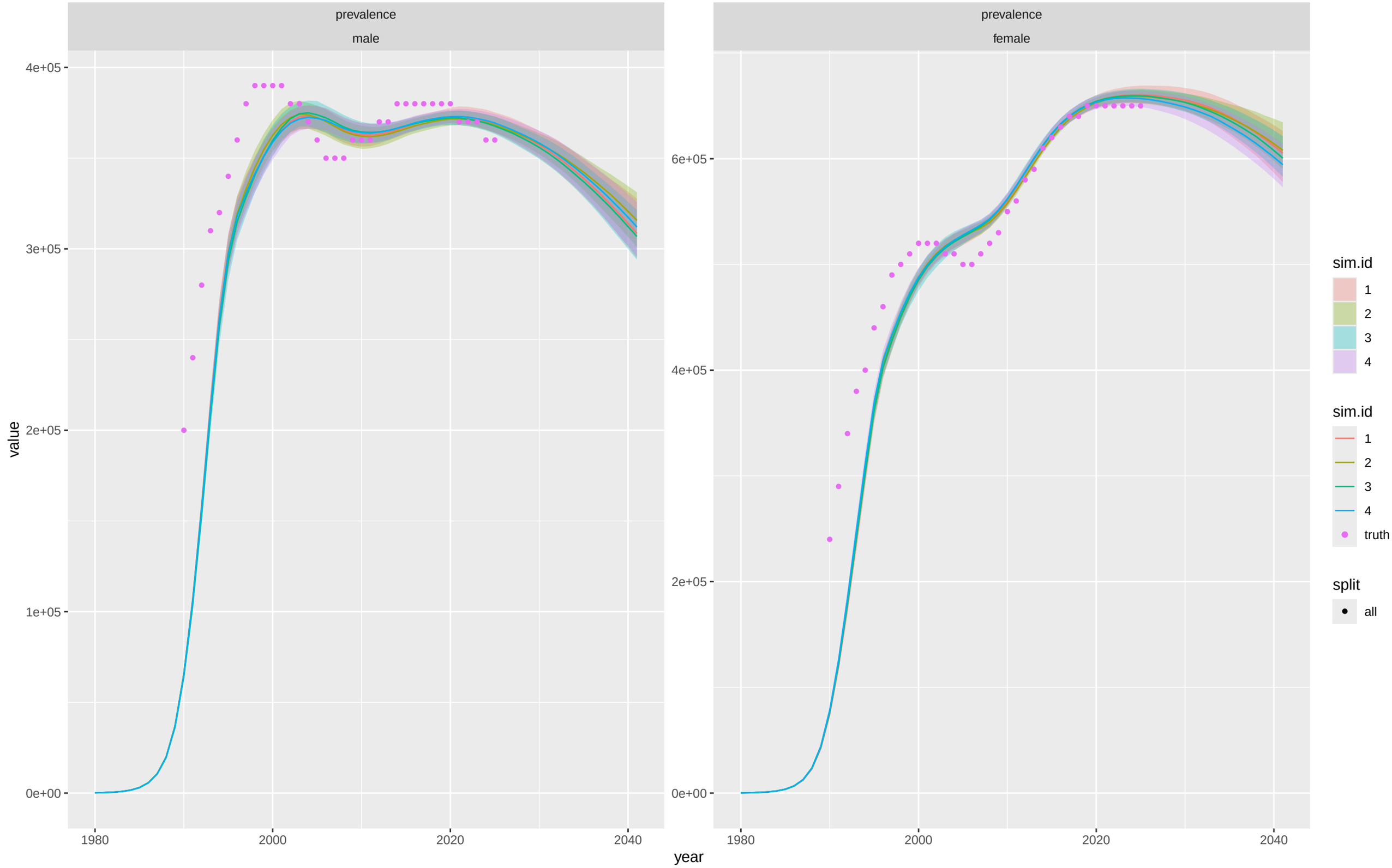
Malawi step 2: cascade.by.age.sex (one color per chain)



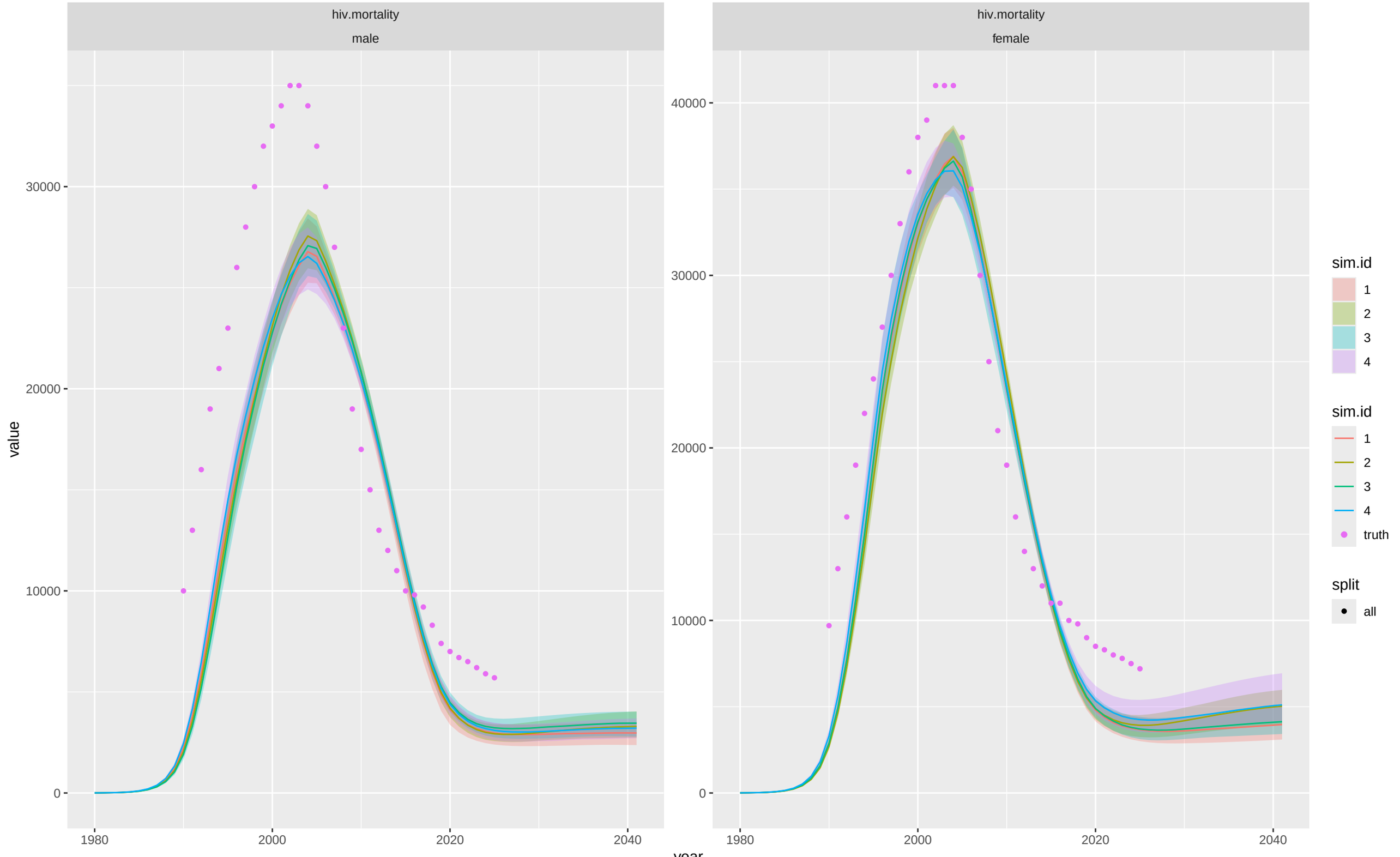
Malawi step 2: incidence.by.sex (one color per chain)



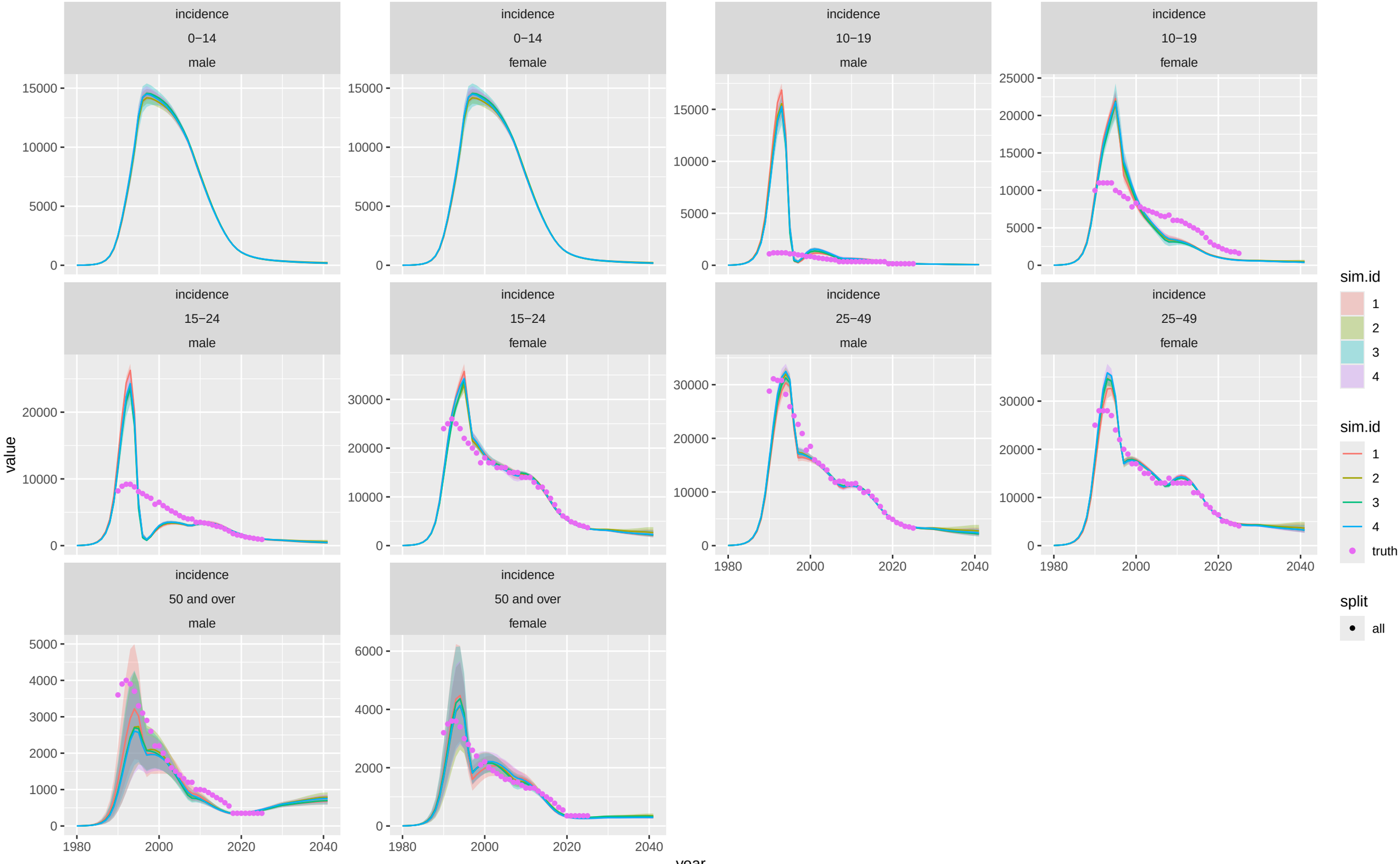
Malawi step 2: prevalence.by.sex (one color per chain)



Malawi step 2: hiv.mortality.by.sex (one color per chain)



Malawi step 2: incidence.by.age.sex (one color per chain)



Malawi step 2: prevalence.by.age.sex (one color per chain)

prevalence
0–14
male

prevalence
0–14
female

prevalence
10–19
male

prevalence
10–19
female

prevalence
15–24
male

prevalence
15–24
female

prevalence
25–49
male

prevalence
25–49
female

prevalence
50 and over
male

prevalence
50 and over
female

value

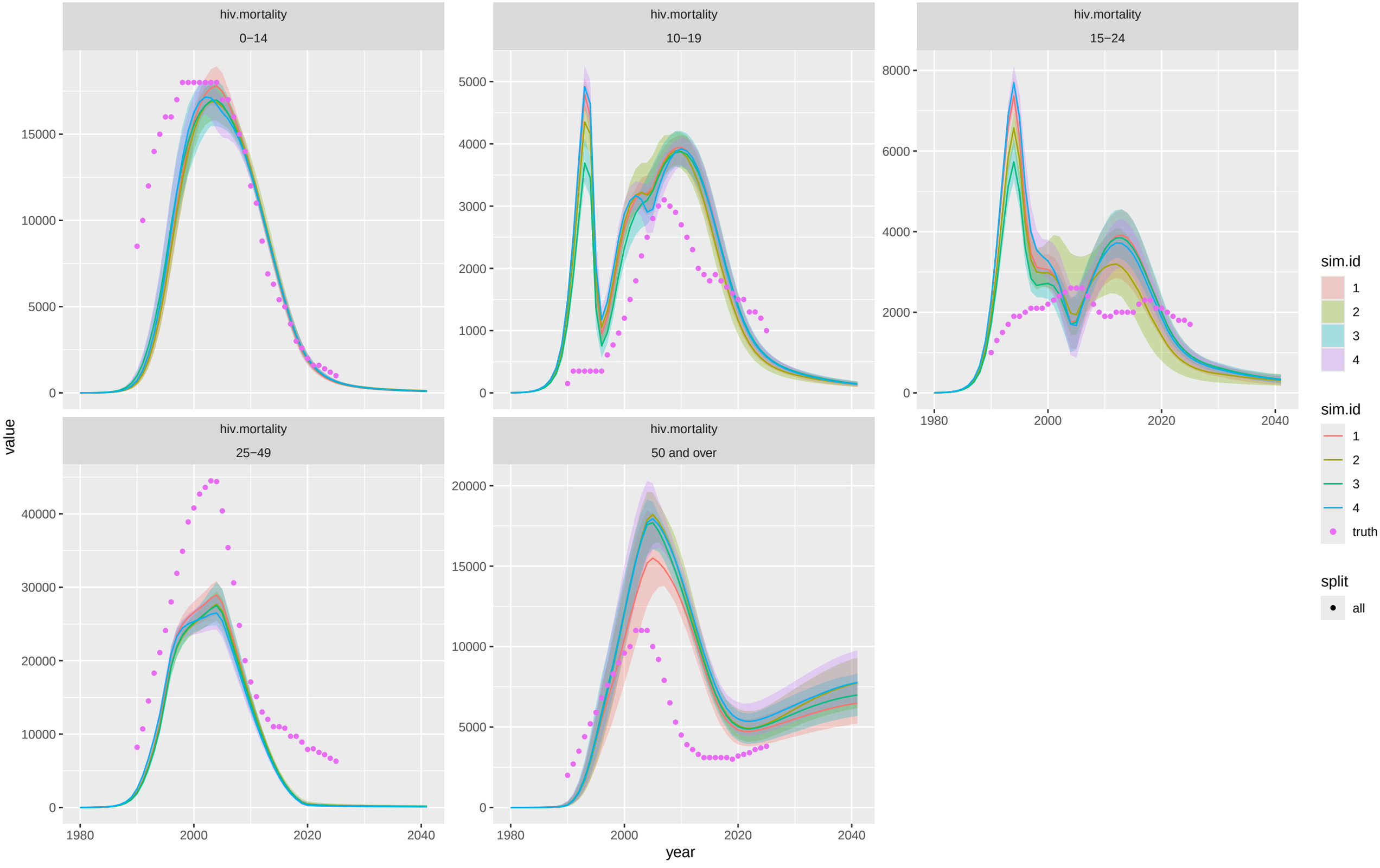
year

sim.id
1
2
3
4

sim.id
1
2
3
4
truth

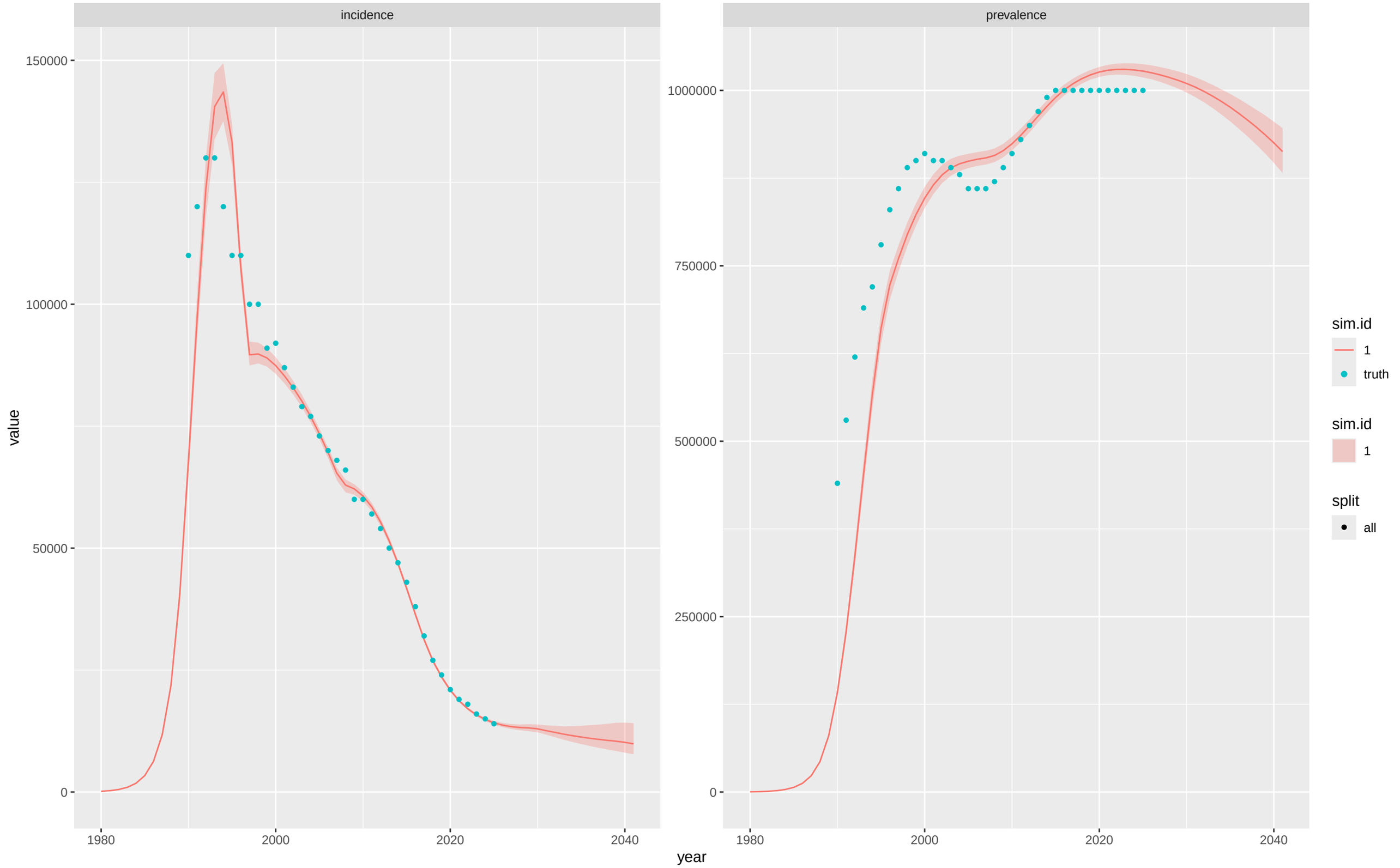
split
all

Malawi step 2: hiv.mortality.by.age (one color per chain)

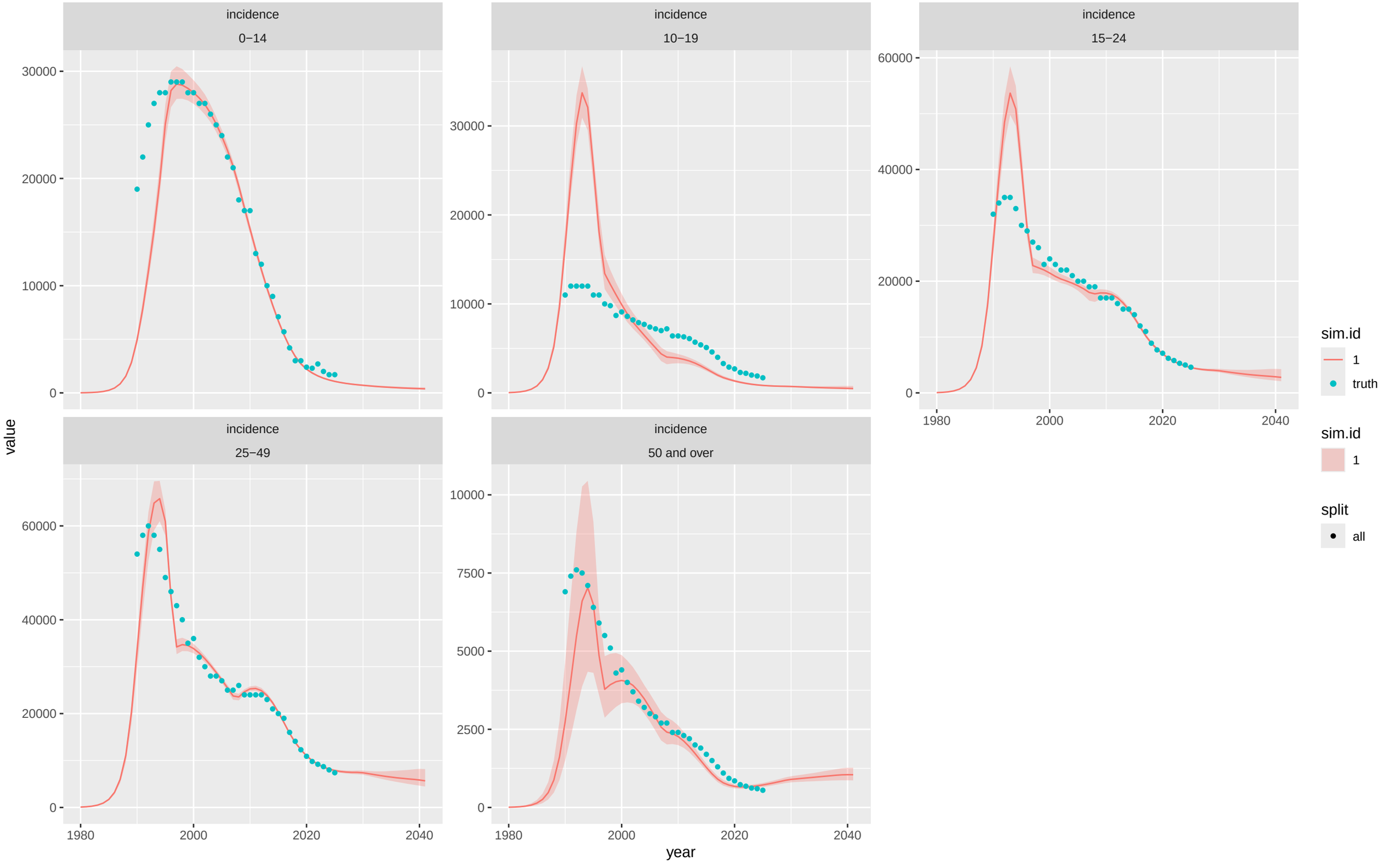


Malawi step 2: hiv.mortality.by.age.sex (one color per chain)

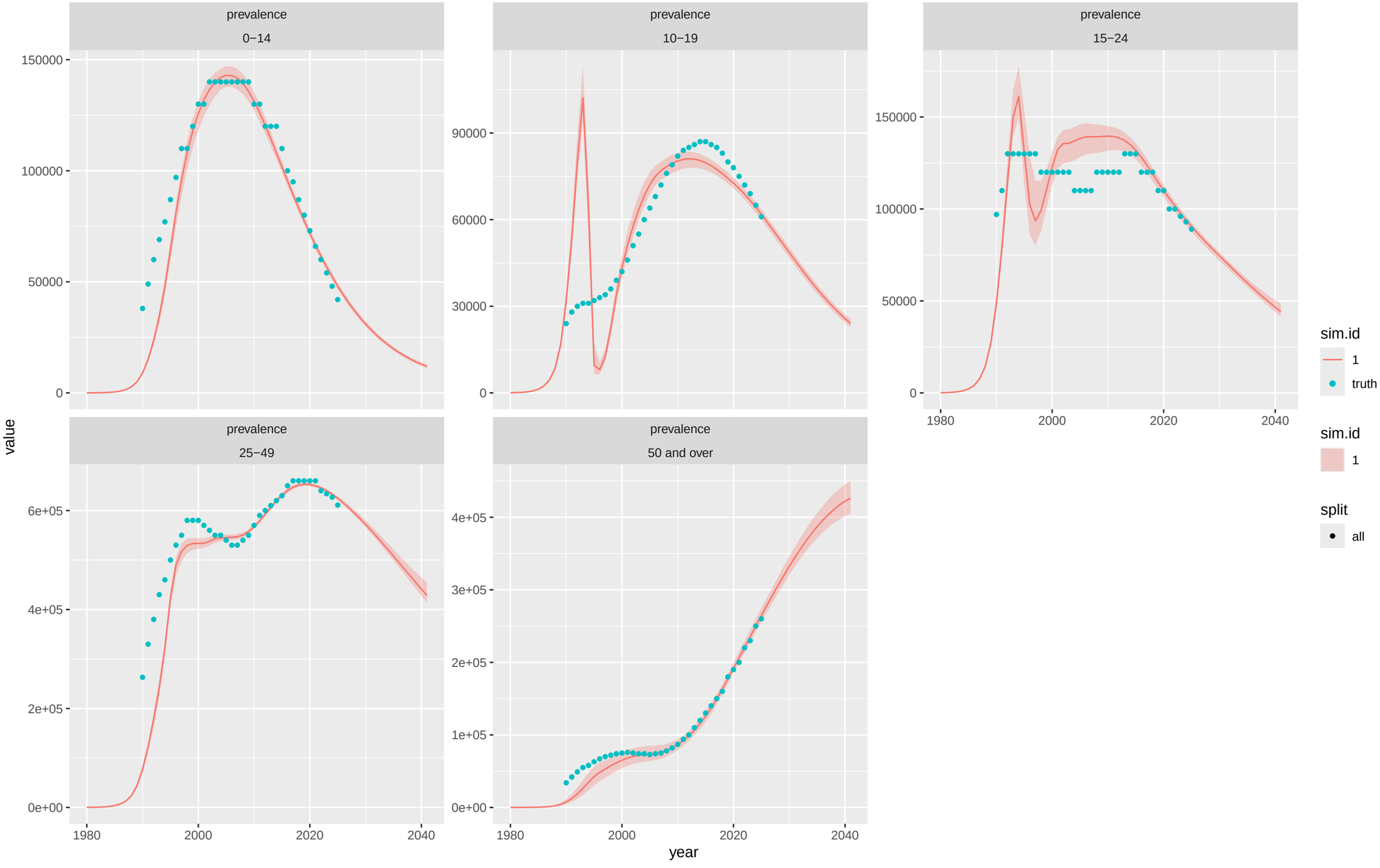
Malawi: fit (merged Step 2 simset)



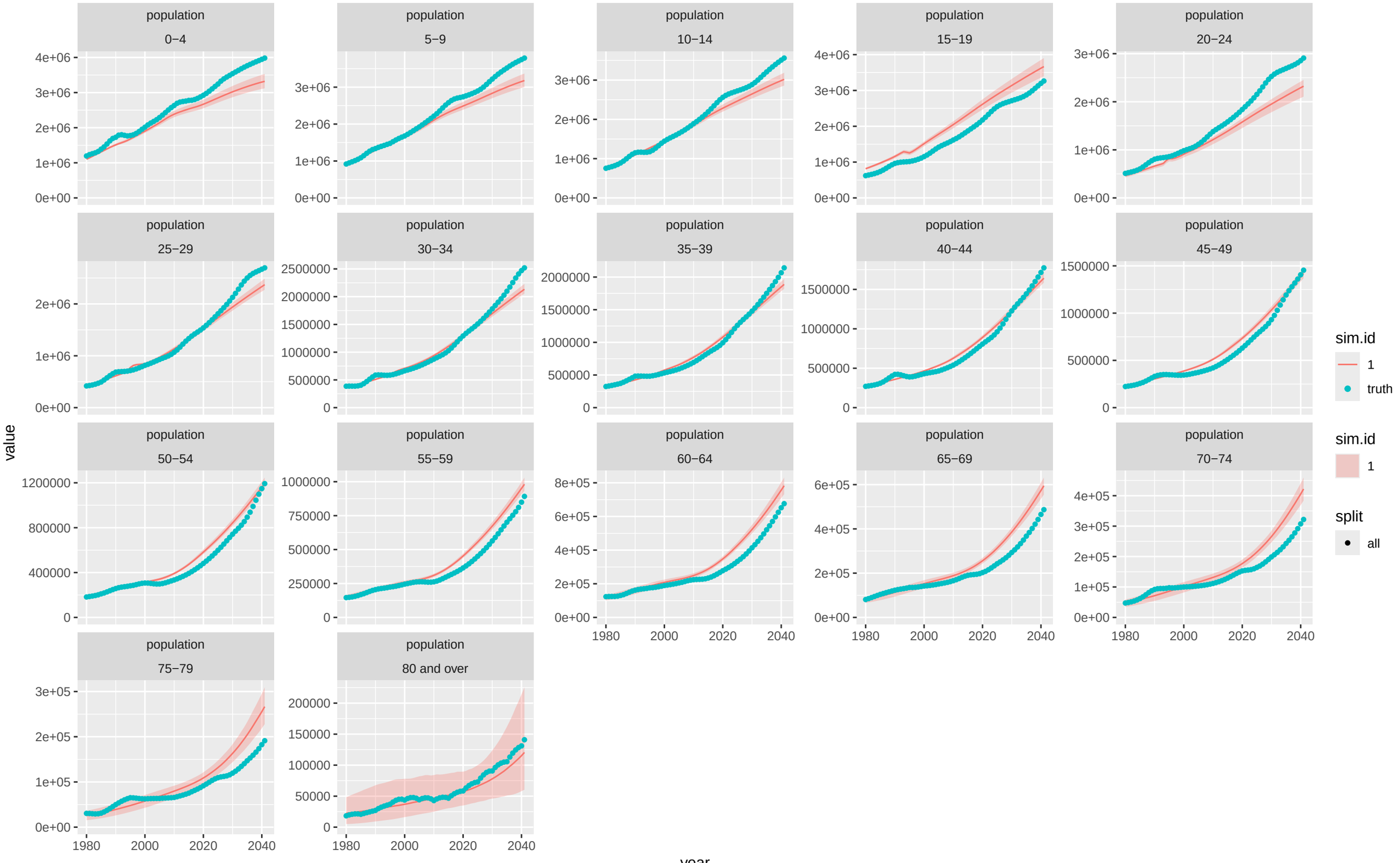
Malawi: incidence.by.age (merged Step 2 simset)



Malawi: prevalence.by.age (merged Step 2 simset)



Malawi: population.by.age (merged Step 2 simset)



Malawi: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

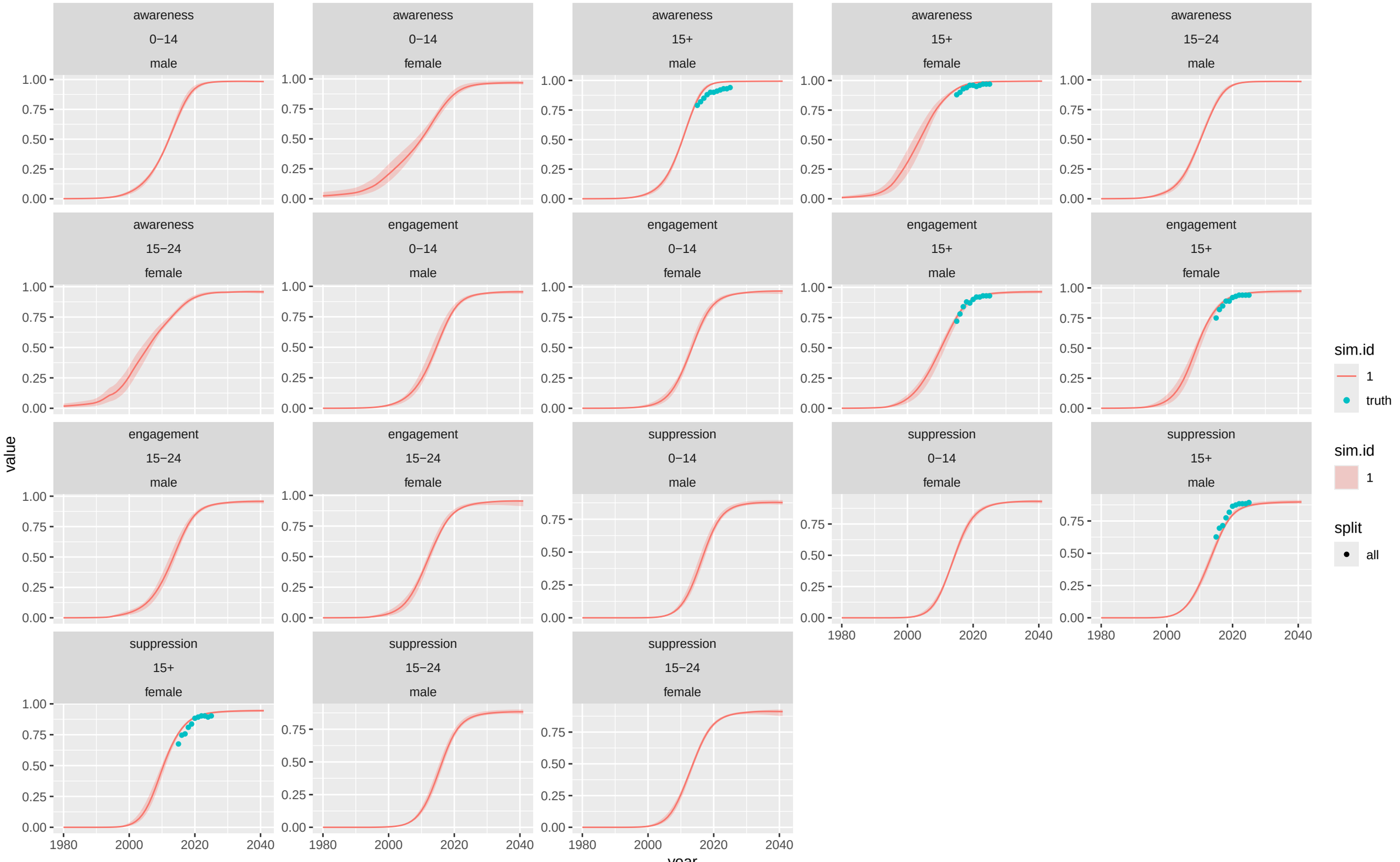
sim.id

1

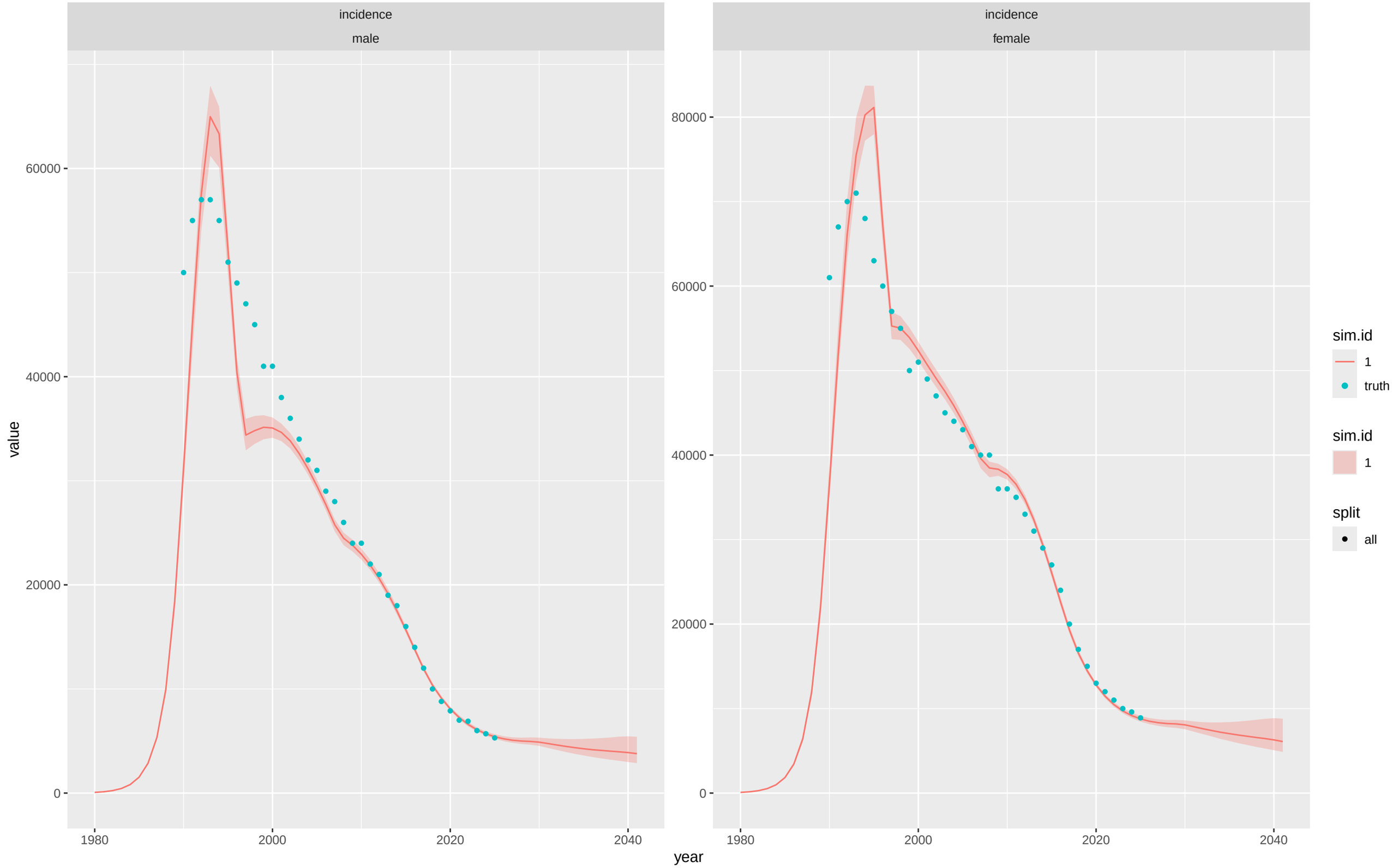
split

all

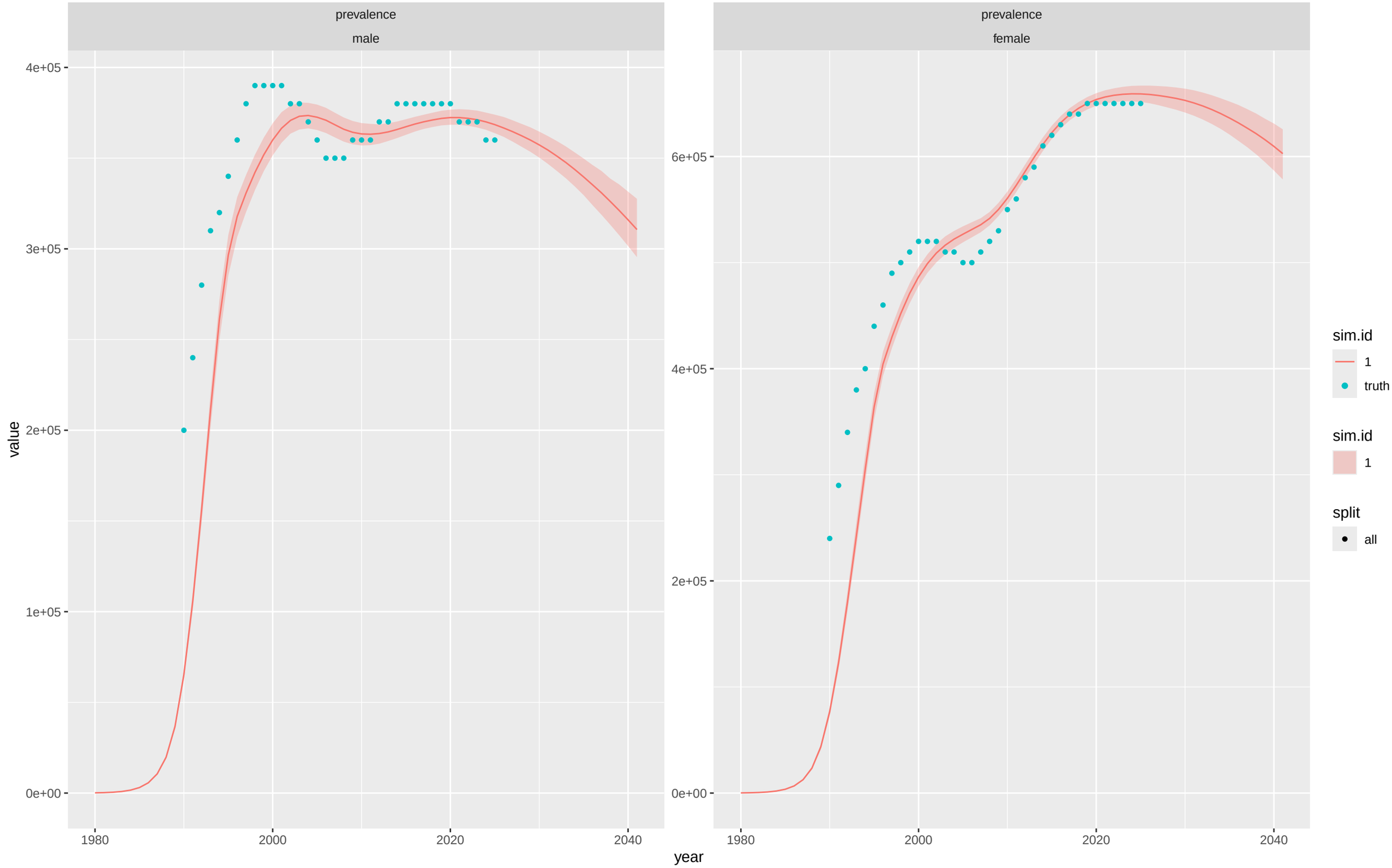
Malawi: cascade.by.age.sex (merged Step 2 simset)



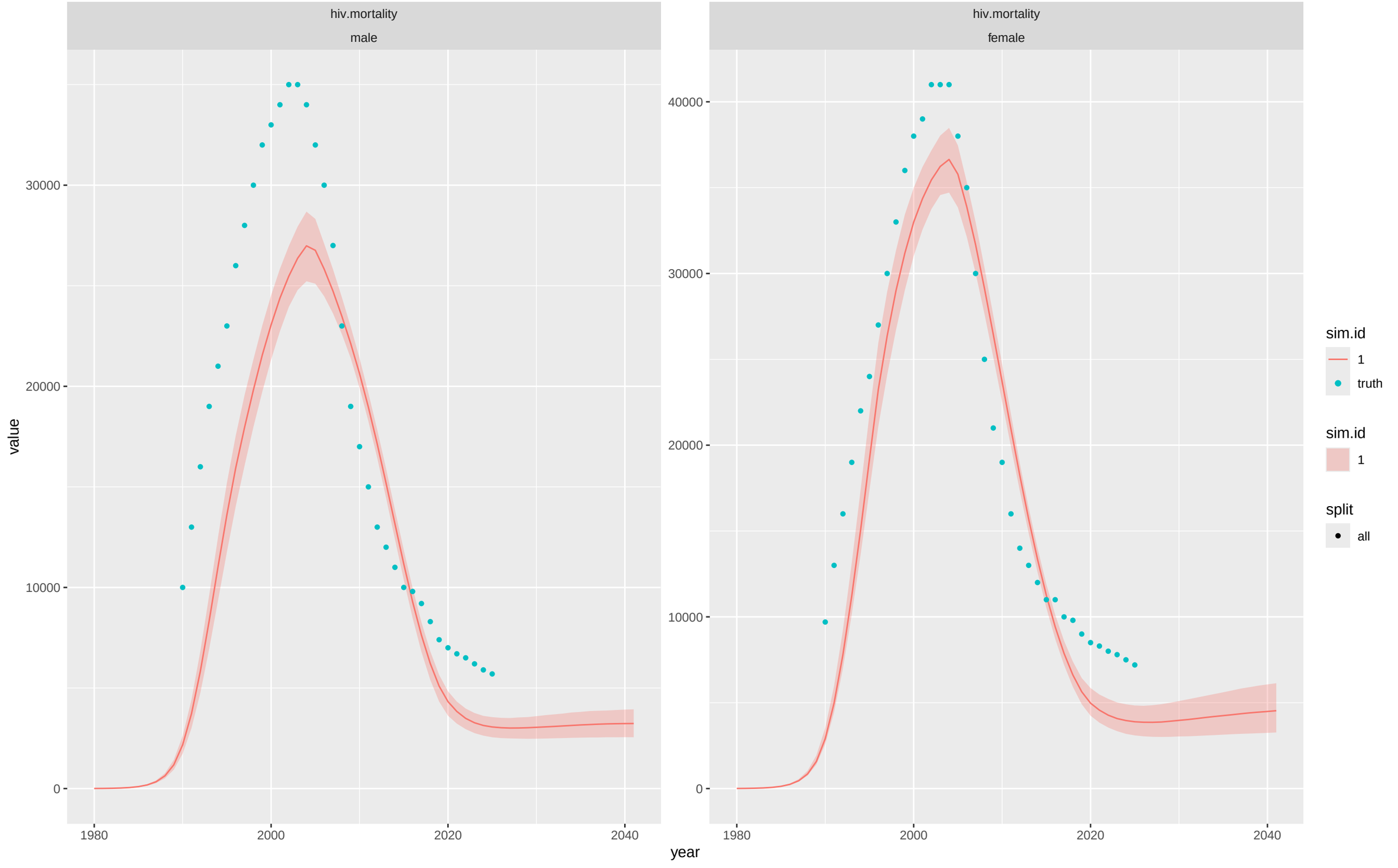
Malawi: incidence.by.sex (merged Step 2 simset)



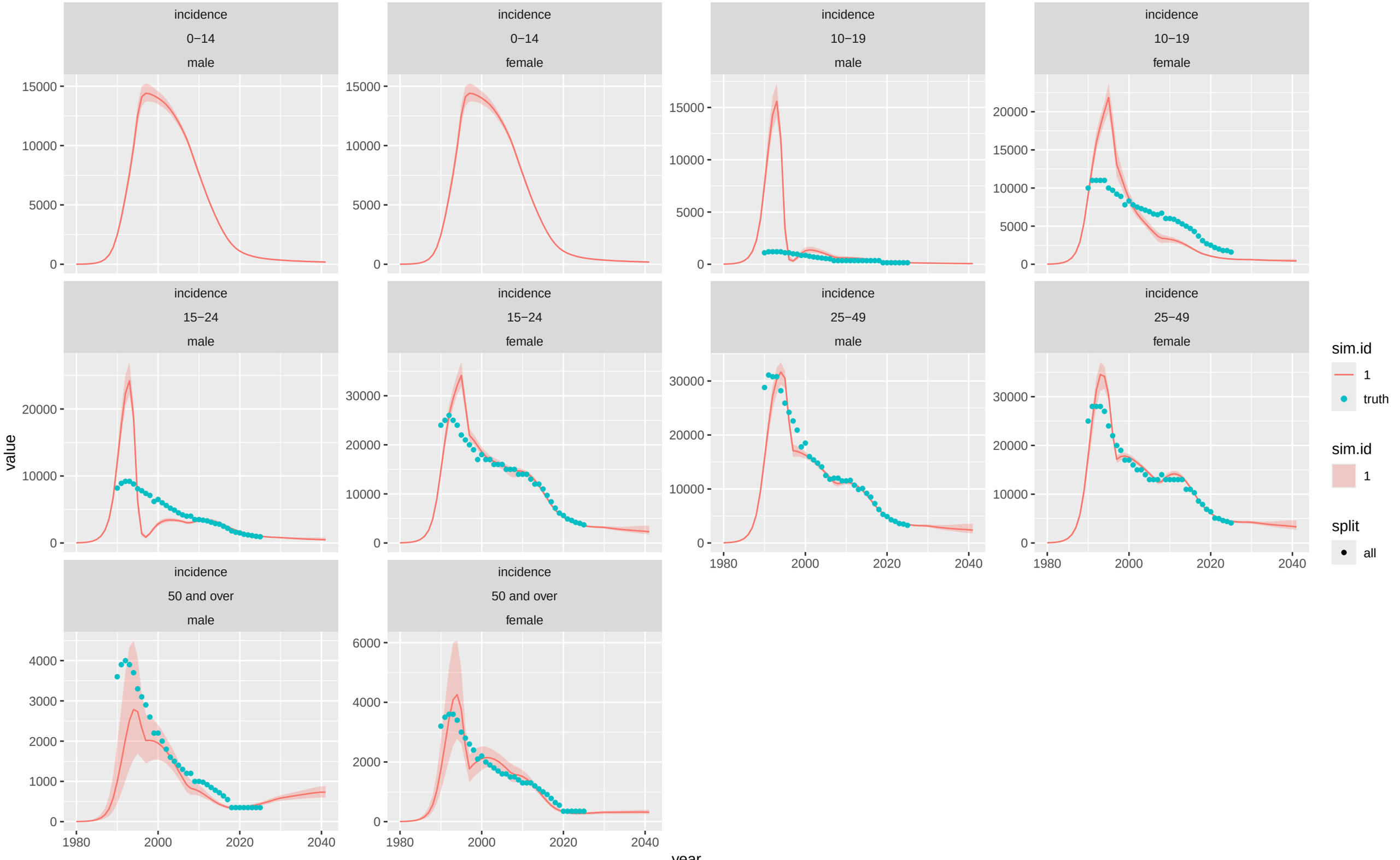
Malawi: prevalence.by.sex (merged Step 2 simset)



Malawi: hiv.mortality.by.sex (merged Step 2 simset)



Malawi: incidence.by.age.sex (merged Step 2 simset)



Malawi: prevalence.by.age.sex (merged Step 2 simset)

prevalence 0–14 male

prevalence 0–14 female

prevalence 10–19 male

prevalence 10–19 female

prevalence 15–24 male

prevalence 15–24 female

prevalence 25–49 male

prevalence 25–49 female

prevalence 50 and over male

prevalence 50 and over female

value

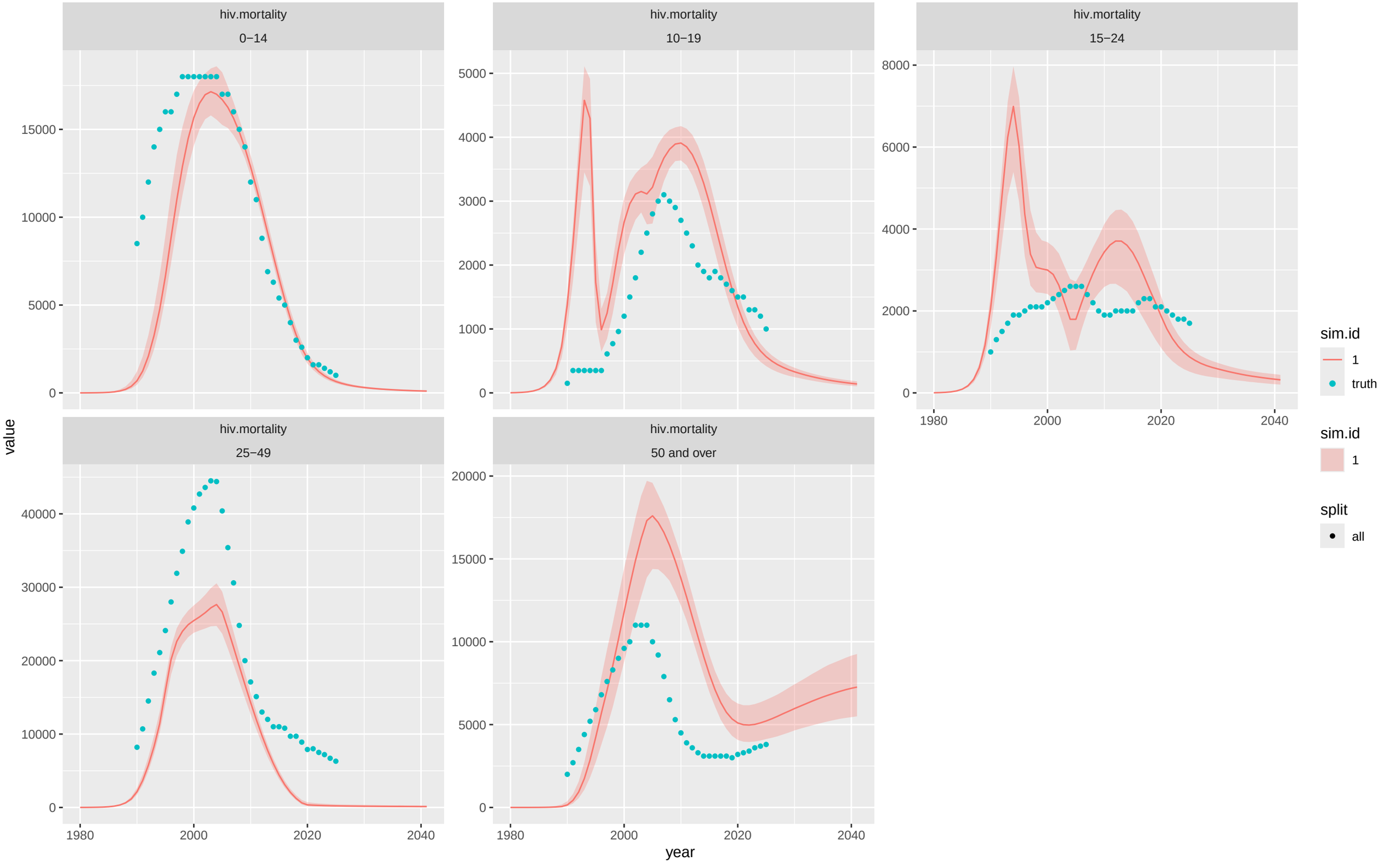
year

sim.id
1
truth

sim.id
1

split
all

Malawi: hiv.mortality.by.age (merged Step 2 simset)



Malawi: hiv.mortality.by.age.sex (merged Step 2 simset)

